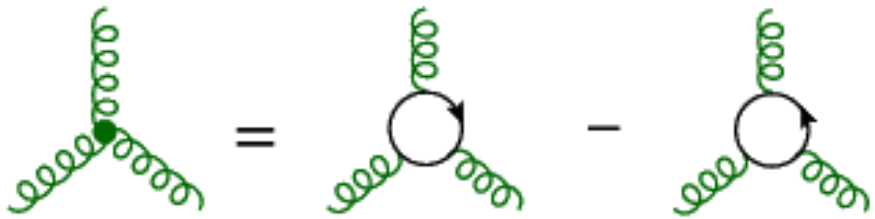
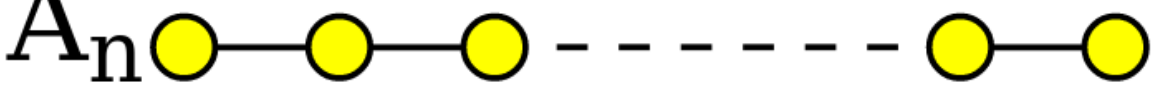
$$\begin{pmatrix} 2 & -1 & 0 & \dots & 0 \\ -1 & 2 & -1 & \dots & 0 \\ 0 & -1 & 2 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 2 \end{pmatrix}$$



# A Ridiculous Formula for the Number of SU(N) Gluonic Color Factors

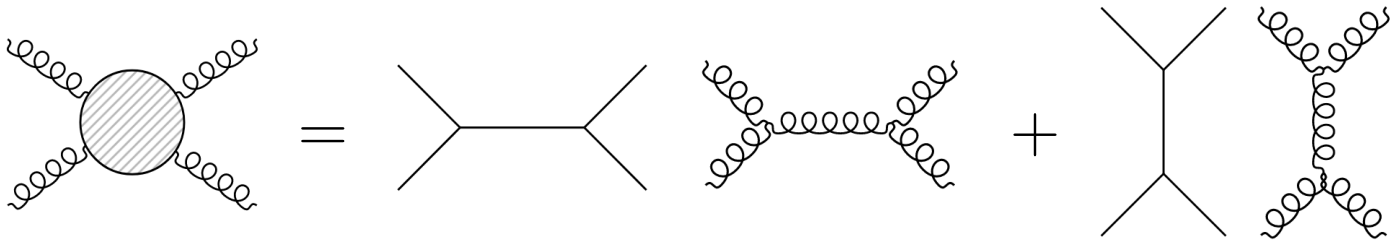
Presented by Michael Plesser
From 24XX.XXXX with J. Bourjaily and C. Vergu






# Color Factors in Gauge Theory

In gauge theory we can organize amplitudes into color factors $\mathcal{C}_i$ and partial amplitudes $A_i$



The $\mathcal{C}_i$ satisfy (group-specific) identities, EG

$$\text{SU(2):} \quad \text{Tr}[123] + \text{Tr}[132] = 0 \leftrightarrow d^{abc} = 0$$

$$\text{SU(3):} \quad \text{Tr}[123] + \text{Tr}[132] \neq 0 \leftrightarrow d^{abc} \neq 0$$

A natural question:

How many independent n-pt color factors are there for a given group?

# Counting with Schur's Lemma

**The TI;dr**

The multiplicity of $\mathbf{1}$ in $\mathbf{Ad}^{\otimes n}$ gives the number of color factors for n-external gluons, $|\mathcal{C}_n|$, to all loop orders

SU(N) n-pts:

$$|\mathcal{C}_n| = \langle \mathbf{1} | \mathbf{Ad}^{\otimes n} \rangle$$

A bound is given by the *Subfactorial* or *Derangement* numbers

$$|\mathcal{C}_n| \leq \ !n$$

This bound is saturated when $n \leq N$

$$\boxed{|\mathcal{C}_{n \leq N}| = \ !n \ = \left[\frac{n!}{e}\right]}$$

(With $[x]$ meaning 'x rounded to the nearest integer')



# Embracing the Derangement!

Derangements are permutations with no 1-cycles
Wrapping each cycle with 'Tr' gives the Deranged trace basis

EG $n = 4$:

$$\left[\frac{4!}{e}\right] = [8.829...] \implies |\mathcal{C}_4| \leq 9$$

| | | |
|---|---|---|
| Tr[1234] | Tr[1342] | Tr[1423] |
| Tr[1243] | Tr[1324] | Tr[1432] |
| Tr[12]Tr[34] | Tr[13]Tr[24] | Tr[14]Tr[23] |

These satisfy 6 SU(2) identities,
1 SU(3) identity, and
0 SU($N \geq 4$) identities



# More to the Story!

- Tree-level "irrep basis" is smaller than DDM
- Cayley-Hamilton Identities
  (Exponential growth for large n, not factorial!)
- Other gauge groups
  ($E_8$ is "simpler" than SU(3)??? $D_n$ does what???)
- ST duality and graphical reduction
  (Biedenharn-Elliot identity, hexagon relation, etc)
- Motzkin and Riordan numbers for SU(2)
  (Racah W's and Wigner 6j's)
- Adding fundamentals and higher reps, its easy!

If any of this sounds cool, let's talk
(or email mkp5771@psu.edu)

Thanks!

